

Diurnal changes in pore water sulfide concentrations in the seagrass *Thalassia testudinum* beds: the effects of seagrasses on sulfide dynamics

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Abstract

The dynamics of the seagrass–sulfide interaction were examined in relation to diel changes in sediment pore water sulfide concentrations in *Thalassia testudinum* beds and adjacent bare areas in Corpus Christi Bay and lower Laguna Madre, Texas, USA, during July 1996. Pore water sulfide concentrations in seagrass beds were significantly higher than in adjacent bare areas and showed strong diurnal variations; levels significantly decreased during mid-day at shallow sediment depths (0–10 cm) containing high below-ground tissue biomass and surface area. In contrast, diurnal variations in sediment sulfide concentrations were absent in adjacent bare patches, and at deeper (> 10 cm) sediment depths characterized by low below-ground plant biomass or when the grasses were experimentally shaded. These observations suggest that the mid-day depressions in sulfide levels are linked to the transport of photosynthetically produced oxygen to seagrass below-ground tissues that fuels sediment sulfide oxidation. Lower sulfide concentrations in bare areas are likely a result of low sulfate reduction rates due to low organic matter available for remineralization. Further, high reoxidation rates due to rapid exchange between anoxic pore water and oxic overlying water are probably stimulated in bare areas by higher current velocity on the sediment surface than in seagrass beds. The dynamics of pore water sulfides in seagrass beds suggest no toxic sulfide intrusion into below-ground tissues during photosynthetic periods and demonstrate that the sediment chemical environment is considerably modified by seagrasses. The reduced sediment sulfide levels in seagrass beds during photosynthetic periods will enhance seagrass production through reduced sulfide toxicity to seagrasses and sediment microorganisms related to the nutrient cycling. © 2000 Elsevier Science B.V. All rights reserved.

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1. Introduction

Seagrasses often grow in oxygen-free, strongly reduced sediments. In anoxic coastal marine sediments, sulfate reduction is the main terminal process in the mineralization of organic matter (Jørgensen, 1977a,c; Hines and Lyons, 1982; Thode-Andersen and Jørgensen, 1989). Up to half of the deposited organic matter in coastal sediment has been found to be mineralized anaerobically via sulfate reduction (Jørgensen, 1977c, 1982). This remineralization process supplies nutrients for seagrass growth but results in accumulation of highly toxic compounds, particularly hydrogen sulfides. Sulfides can be trapped in the sediment by precipitation with metal ions, but some remain dissolved in sediment pore water (Thode-Andersen and Jørgensen, 1989). Toxic sulfides have been reported as a causal factor in low productivity and in several physiological disorders of plants living in submerged or flooded anoxic sediments (Allam and Hollis, 1972; Joshi and Hollis, 1977; Pulich, 1983; Ingold and Havill, 1984; Koch and Mendelssohn, 1989; Koch et al., 1990; Bagarinao, 1992; Van Wijck et al., 1992; Goodman et al., 1995).

In Florida Bay, pore water sulfide concentrations were considerably higher in die-off areas than in healthy seagrass beds, suggesting that sulfide toxicity may have played a role in seagrass loss (Carlson et al., 1994). Decreases in maximum photosynthetic rate ($P_{\max}$) of *Zostera marina* have been reported in response to increases in sediment sulfide concentrations (Goodman et al., 1995). Therefore, elevated sediment sulfide levels may contribute to seagrass die-off in areas which have low underwater light availability due to heavy epiphyte fouling or high water column turbidity. Since sulfides have harmful impacts on seagrass energy metabolism and photosynthesis, the dynamics of pore water sulfides in seagrass beds play a critical role in the regulation of productivity.

Seagrasses have adapted to reducing conditions and sulfide toxicity with a well-developed internal lacunae system (Roberts et al., 1984; Roberts and Caperon, 1986), which allows transport of oxygen from photosynthetic tissues to below-ground tissues. Oxygen transport to the root–rhizome system is linked to photosynthetic oxygen production (Smith et al., 1984), and transported oxygen can support aerobic respiration in root tissues (Sand-Jensen et al., 1982; Smith et al., 1984; Caffrey and Kemp, 1991). Sulfides are spontaneously oxidized in the presence of oxygen, and the oxidation can be accelerated by enzymatic or non-enzymatic catalysis in plant root tips (Lee et al., 1999). Therefore, oxygen transported in internal spaces of below-ground tissues and in the surrounding rhizosphere limits sulfide intrusion into below-ground tissues.

Frequently, oxygen transport exceeds below-ground tissue's respiratory requirements, and excess oxygen is released into the sediments (Oremland and Taylor, 1977; Pedersen et al., 1998). Released oxygen in the rhizosphere oxidizes reduced substances (e.g., sulfide) in the close vicinity of the roots. Consequently, diurnal fluctuations of pore water sulfide levels are apparent in seagrass beds. However, few studies have addressed temporal or diurnal variations in pore water sulfide concentrations in seagrass beds. In

this study, the dynamics of pore water sulfides were examined in *Thalassia testudinum* beds and adjacent bare areas in lower Laguna Madre and Corpus Christi Bay, Texas, USA. We measured diurnal changes in sulfide concentrations under natural conditions and experimentally shaded beds to determine the effects of photosynthesis on daily sulfide variations. Diurnal changes in sulfide concentrations with sediment depth were correlated with depth profiles of below-ground tissues, sediment porosity, and sediment organic carbon (C) and nitrogen (N) content in conjunction with continuous measurement of underwater quantum irradiance.

2. Materials and methods

2.1. Study sites

Dynamics of pore water sulfide concentrations were examined at sites located on the eastern side of Corpus Christi Bay (27°49'N, 97°07'W) and the lower Laguna Madre (26°08'N, 97°12'W) during July 1996. *Thalassia testudinum*, *Halodule wrightii* and *Syringodium filiforme* are the dominant seagrass species at these sites. This study was conducted in monotypic meadows of *T. testudinum* and adjacent bare areas at an average water depth of 0.8 m in Corpus Christi Bay and 1.2 m in lower Laguna Madre.

2.2. Photon flux measurement

Photosynthetically active radiation (PAR, 400–700 nm) was collected continuously using a LI-193SA spherical quantum sensor at canopy level, which provides input to a LI-1000 datalogger (LI-COR, Lincoln, NE) during the duration of the experiments. Some seagrass beds (4 × 4 m) were shaded during the experiment by neutral density shade cloth to examine the effects of seagrass photosynthesis on dynamics of pore water sulfide concentration. On the evening prior to diel sampling of sulfides, shade cloth was deployed over the experimental area. Photon flux density (PFD, $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$) was measured at 5-s intervals and averaged for 5 min.

2.3. Sediment organic C and total N

For determination of porosity, organic C and total N, four replicate sediment cores were collected from seagrass beds and adjacent bare areas using a 9-cm diameter corer. Within a sampling site, replicate samples for sediment cores were at least 3 m apart. Cores from bare sites were taken at least 3 m distant from seagrass plants to minimize the effects of the seagrasses on sediment geochemistry. Cores were sliced at 3-cm depth intervals to examine chemical and physical depth profiles. Wet samples were homogenized and a subsample was weighed into a preweighed 20-ml glass scintillation vial. The sample was dried at 60°C to a constant weight and the percent of water in sediment (P_w) was calculated by subtraction of dry weight (W_o) from wet weight (Hedges and Stern, 1984). Sediment porosity (ϕ) was calculated from the P_w and sediment grain density (D_g) using the following equation:

$$\phi = P_w / \{(100 - P_w) / D_g + P_w\}$$

D_g was calculated by measuring weight and volume of the sediment.

Sediment organic C and total N were measured using an elemental analyzer (Carlo Erba EA 1108). To remove inorganic C, the dried samples were treated with 5 ml of 1 N HCl and held at room temperature for 2 h. After redrying at 60°C, the dried sample was allowed to sit open for at least 24 h to absorb water vapor from the air. After equilibration, the sample was reweighed to obtain the final weight (W_f). The sample was homogenized and 10–30-mg subsamples were weighed for elemental analysis. The gravimetric ratio (W_f/W_o) normalized the sediment organic C and total N contents of the CHN sample back to that of the original sample (Froelich, 1980; Hedges and Stern, 1984).

2.4. Pore water sulfide

Sediment pore water samples were collected with a modified pore water sampler similar to Zimmermann et al. (1978) under anaerobic conditions. Pore water samplers were constructed from 1.2-cm diameter glass tubing with plastic tips which were cotton filled and had holes punched through the sides. Samplers were purged with nitrogen gas and inserted in the sediment, and a vacuum was then placed on the sampler using a 50-ml syringe. Pore water was collected through the sample tubing for about 20 min after placing a vacuum on the sampler. Dissolved sulfide content of pore water was determined colorimetrically according to Cline (1969). A 5-ml pore water sample was fixed in the field by adding 0.4 ml of mixed diamine reagent. Color development was allowed to proceed in the dark, after which the absorbance was determined spectrophotometrically at 670 nm. Dilutions were made after color development with distilled water. The concentrations of sulfide in the samples were calculated by comparison to a standard curve of known sulfide concentrations.

To examine the diurnal changes in pore water sulfide concentrations, pore water samples were collected at 3-h intervals from the unshaded and shaded seagrass beds and from adjacent bare areas during an entire day in Corpus Christi Bay. In lower Laguna Madre, changes in pore water sulfide concentrations were measured in detail during the daytime. Pore water was collected at the sediment depth (8–10 cm) containing maximum below-ground biomass. Diurnal depth profiles of sulfide were analyzed with pore water from 2, 5, 8, 11 and 14 cm sediment depths collected at 08:00–10:00 and at 13:00–15:00 h from the seagrass beds and adjacent bare areas (lower Laguna Madre only).

2.5. Depth profile of below-ground biomass

Four replicate biomass samples from seagrass beds were collected with a 9-cm diameter coring device. The cores were sliced at 3-cm depth intervals to examine the sediment depth profile of below-ground biomass. Samples were thoroughly cleaned of sediments, separated into leaf (sheath), short shoot (vertical rhizome), rhizome and root tissues, and dried at 60°C to a constant weight. Samples were weighed and biomass was

converted to an aerial estimate (g dry wt m^{-2}). Surface areas of below-ground tissues were estimated by measuring the average diameter and total length of subsamples of each tissue. Subsamples were dried and used to convert dry weight to surface area of each tissue.

2.6. Statistics

All values are reported as means $\pm$ 1 S.E. Statistical analyses were performed using a general linear procedure (SAS Institute, 1989). Data were tested for normality and homogeneity of variance to meet the assumptions of parametric statistics, and assumptions were satisfied for all data tested. Significant differences in underwater PFD between plots, and in biomass and tissue surface area among sediment depths were tested using a one-way ANOVA. Significant differences in sediment porosity, organic C and total N between seagrass beds and adjacent bare areas, and among sediment depths were tested using a two-way ANOVA. A two-way ANOVA was also used to test for significant differences in pore water sulfide concentrations among sites and sampling times for diurnal variation, and among sediment depths and sampling times for sediment depth variation. When a significant difference among variables was observed, the means were analyzed with a Tukey multiple comparison test to determine where the significant differences occurred among variables.

3. Results

3.1. Underwater irradiance and sediment chemistry

Underwater PFD exhibited a typical daily sigmoidal curve in both Corpus Christi Bay and lower Laguna Madre sites (Fig. 1). Peak daily levels of underwater PFD ranged from 1650 to 1900 $\mu\text{mol m}^{-2} \text{s}^{-1}$ in lower Laguna Madre and Corpus Christi Bay, respectively, between 13:00 and 14:00 h. Shading by black cloth significantly ($p < 0.001$) decreased underwater PFD (Fig. 1). Within the shaded areas, seagrass received only about half of the daily maximum underwater PFD between 12:00 and 13:30 h.

Sediment organic C and total N content were significantly ($p < 0.001$) higher in seagrass beds than adjacent bare areas (Fig. 2). Average C and N content were 0.78 and 0.092% in seagrass beds and 0.59 and 0.069% in bare areas, respectively. There were significant ($p < 0.001$) changes in C and N content as a function of sediment depth at both sites, with highest C and N content at 3–6 cm sediment depth and lowest at 12–15 cm depth. Average sediment porosity in seagrass beds ($\phi = 0.52$) was significantly ($p < 0.001$) higher than those in bare areas ($\phi = 0.48$). In both seagrass beds and adjacent bare areas, sediment porosity decreased significantly ($p < 0.001$) with sediment depth below 3 cm (Table 1).

3.2. Below-ground tissue biomass and surface area

Below-ground tissue biomass and surface area showed significant ($p < 0.001$)

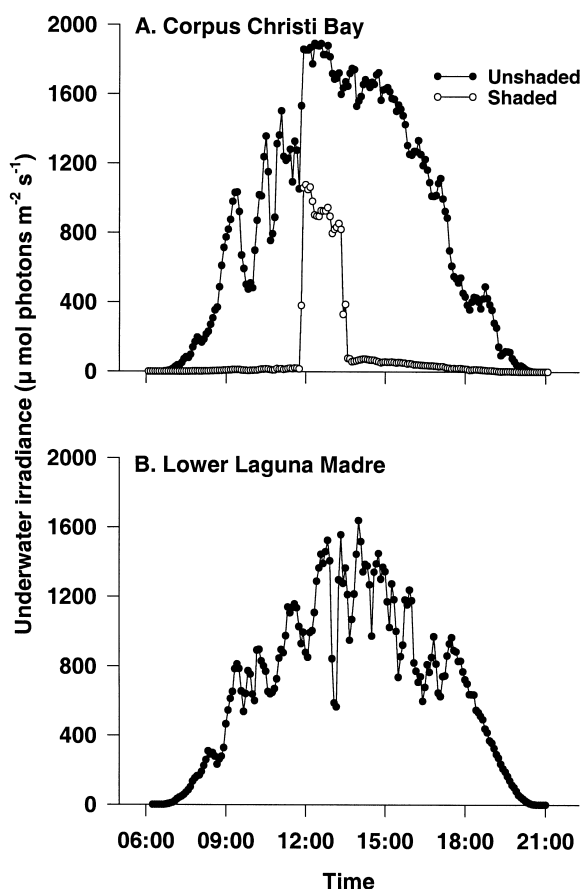


Fig. 1. Underwater photon flux density recorded at seagrass canopy height during July 16, 1996 in Corpus Christi Bay (A) and during July 28, 1996 in lower Laguna Madre (B).

variations with sediment depth (Fig. 3). About 75% of biomass and 65% of total tissue surface area were located between 3 and 9 cm. Over 90% of below-ground tissue biomass and surface areas were present within the top 12 cm of the sediment. Biomass was highest in rhizome tissues, and surface area was greatest in roots. Some live leaf tissues (sheath) were present at the top 6 cm of the sediment.

3.3. Variations in pore water sulfide levels

Pore water sulfide concentrations in *Thalassia testudinum* beds at Corpus Christi Bay were about 160 μM during dark periods, but decreased significantly ($p < 0.01$) to 102 μM during periods of high underwater irradiance (Fig. 4). At the shaded seagrass beds, sulfide concentrations decreased to 139 μM during mid-day, but the decrease was not statistically significant ($p = 0.261$) from the initial dark period. Pore water sulfide

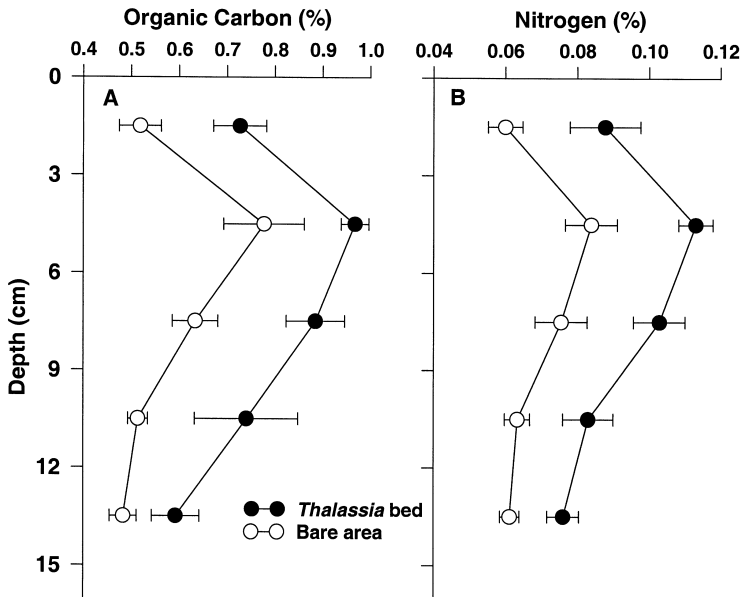


Fig. 2. Sediment organic carbon (A) and total nitrogen content (B) in *Thalassia testudinum* beds and adjacent bare areas in the lower Laguna Madre. Values report mean $\pm$ S.E. ($n=4$).

concentrations in bare areas (75 μM) were significantly ($p<0.001$) lower than those in seagrass beds, and did not show significant ($p=0.988$) diurnal variations (Fig. 4). In lower Laguna Madre, pore water sulfide concentrations in seagrass beds varied inversely with respect to underwater PFD (Fig. 5).

Pore water sulfide concentrations in seagrass beds in lower Laguna Madre were significantly ($p<0.001$) lower during high underwater irradiance periods (13:00–15:00) than during low underwater light periods (08:00–10:00) at sediment depths 0–10 cm (Fig. 6). However, sulfide concentrations in deeper sediments (i.e., 10–15 cm) did not exhibit significant ($p=0.881$) change with sampling time (Fig. 6). Pore water sulfide concentrations in bare areas did not significantly ($p=0.331$) change with sampling time at any sediment depth.

Table 1
Sediment porosity in *Thalassia testudinum* beds and adjacent bare areas in the lower Laguna Madre

Sediment depth (cm)	Porosity	
	<i>Thalassia</i> bed	Bare area
0–3	0.57 $\pm$ 0.02	0.51 $\pm$ 0.02
3–6	0.57 $\pm$ 0.01	0.53 $\pm$ 0.01
6–9	0.53 $\pm$ 0.01	0.48 $\pm$ 0.00
9–12	0.48 $\pm$ 0.01	0.43 $\pm$ 0.01
12–15	0.46 $\pm$ 0.01	0.42 $\pm$ 0.01

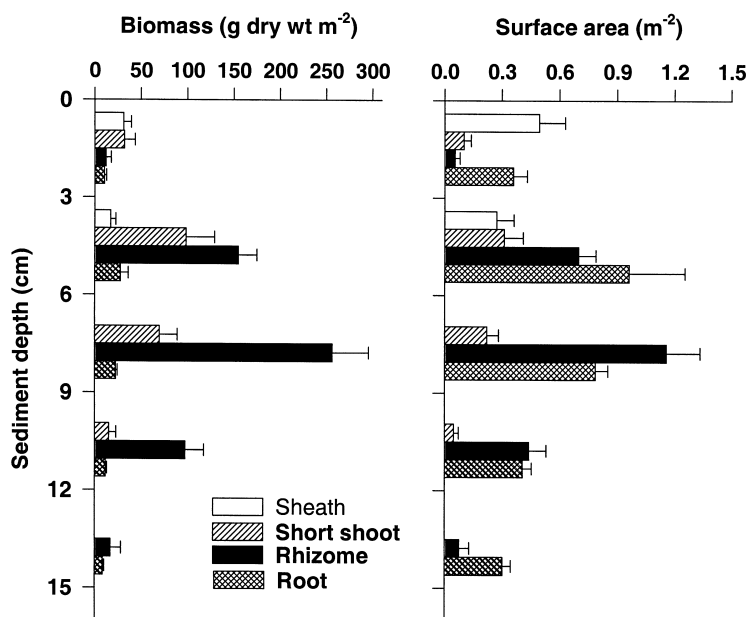


Fig. 3. Vertical distribution of *Thalassia testudinum* below-ground tissues in the lower Laguna Madre. Values report mean±S.E. (n=4).

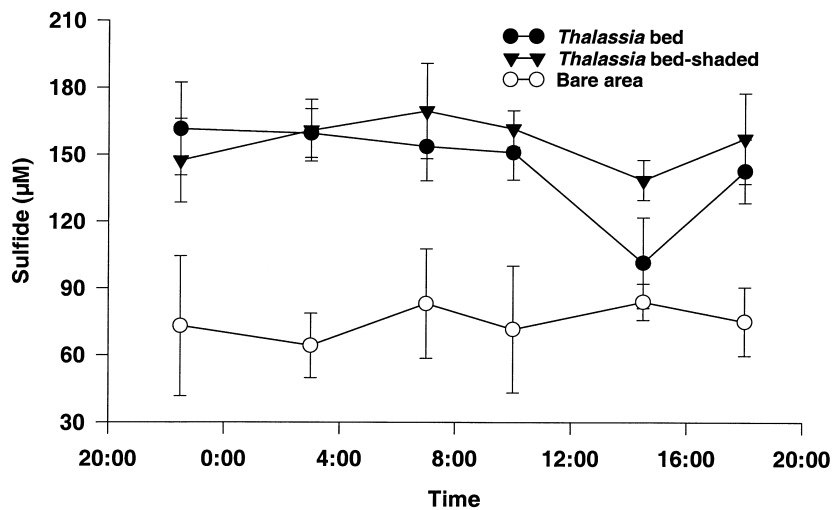


Fig. 4. Diurnal changes of sediment pore water sulfide concentrations in shaded and unshaded *Thalassia testudinum* beds and adjacent bare areas in Corpus Christi Bay. Values report mean±S.E. (n=4–16).

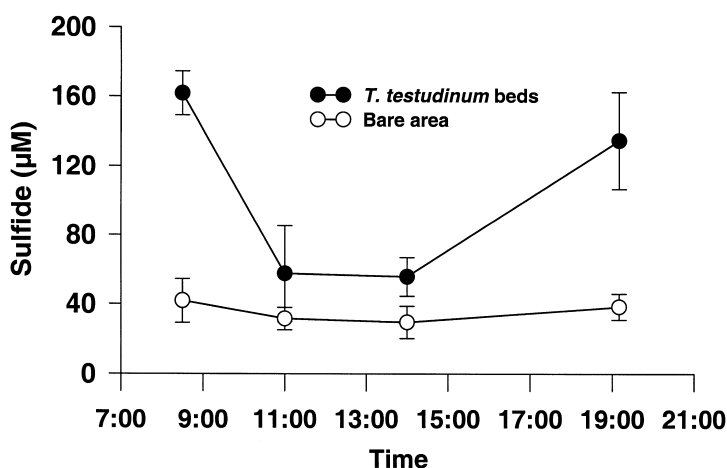


Fig. 5. Changes of sediment pore water sulfide concentrations during daytime in *Thalassia testudinum* beds and adjacent bare areas in lower Laguna Madre. Values report mean $\pm$ S.E. ($n=5-12$).

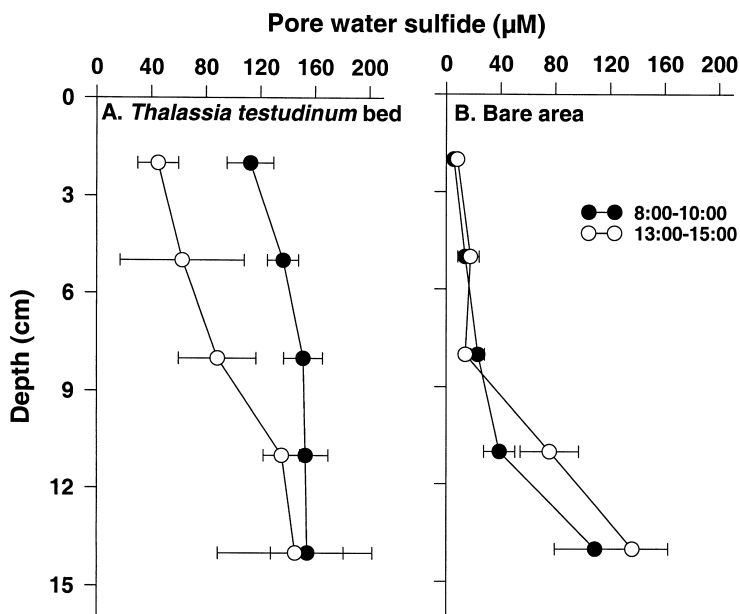


Fig. 6. Changes in sediment pore water sulfide concentrations with sediment depth during low (08:00–10:00) and high (13:00–15:00) underwater light conditions in *Thalassia testudinum* beds (A) and adjacent bare areas (B) in the lower Laguna Madre. Values are mean $\pm$ S.E. ($n=4-12$).

4. Discussion

4.1. Oxidation of pore water sulfide surrounding the rhizosphere

Sediment pore water sulfide concentrations in *Thalassia testudinum* beds decreased significantly during periods of high underwater PFD, coincident with high photosynthetic rates of leaf tissues (Dunton and Tomasko, 1994; Dunton, 1996; Herzka and Dunton, 1997). Pedersen et al. (1998) directly measured the distribution of free oxygen around roots of the seagrass *Cymodocea rotundata* using oxygen microelectrodes and demonstrated that the oxic microzone around roots was thicker in daylight than under darkness. In *T. testudinum* beds, diurnal fluctuation of oxygen concentrations has been reported in gas bubbles from sediments; oxygen content was lowest in early morning, increased during the day, and decreased again by evening (Oremland and Taylor, 1977). In the present study, sediment pore water sulfide concentrations in *T. testudinum* beds were diametrically opposed to the reported temporal oxygen variation.

No significant decrease in pore water sulfide concentrations occurred in the rhizosphere of shaded seagrass beds, where plants received only half of the daily maximum underwater PFD (ca. 800 $\mu\text{mol m}^{-2} \text{s}^{-1}$) for less than 2 h. Additionally, pore water sulfide concentrations in adjacent bare areas did not show any diurnal variations. Therefore, decreases of pore water sulfide concentrations in *T. testudinum* beds during mid-day were probably caused by oxidation of sulfides in the vicinity of seagrass below-ground tissues via transported oxygen which was produced by seagrass photosynthesis. Oxygen transport from photosynthetic tissues to below-ground tissues via internal lacunae system have been well documented in submerged aquatic plants (Penhale and Wetzel, 1983; Smith et al., 1984; Sorrell and Dromgoole, 1987, 1988; Caffrey and Kemp, 1991; Flessa, 1994; Sorrell, 1994).

Sulfides can be oxidized with bacterial mediation (Jørgensen, 1977b; Fenchel and Bernard, 1995). Slow spontaneous sulfide oxidation by oxygen will be accelerated by bacterial intervention. Marine *Beggiatoa* spp. at oxic–anoxic interface have been shown to oxidize hydrogen sulfide at rates which are several thousand times greater than the rates of spontaneous chemical oxidation (Jørgensen and Revsbech, 1983; Nelson et al., 1986). In part, decreases in pore water sulfide concentrations during high photosynthetic periods in the present study are probably a result of sulfide oxidation mediated by the sulfur bacteria. However, the sulfide oxidizing colorless sulfur bacteria (c.a. *Beggiatoa* spp. and *Thiobacillus* spp.) require oxygen or other oxidant for the sulfide oxidation process (Jørgensen, 1977b; Gundersen et al., 1992; Fenchel and Bernard, 1995; Visser et al., 1997). Oxygen supply into sediments of seagrass beds is mainly a consequence of seagrass photosynthesis (Smith et al., 1984; Caffrey and Kemp, 1991). Thus, any bacterial mediated sulfide oxidation in seagrass bed sediments is enhanced by oxygen release from root and rhizome tissues.

4.2. Limit of toxic sulfide intrusion into below-ground tissues

When photosynthetically produced oxygen is released into the sediment and creates a thin oxic layer around root tissues, sulfides are unable to diffuse into seagrass below-

ground tissues. Therefore, toxic sulfide intrusion into below-ground tissues will normally be limited during daytime hours. In the dark, seagrass below-ground tissues rely on oxygen stored in the lacunae systems from day-time photosynthesis to sustain root aerobic respiration and to prevent toxic sulfide intrusion. Oxygen diffused out of the *Ruppia maritima* root tissues in the dark, when plants were given a light pretreatment (Thursby, 1984). Pedersen et al. (1998) have also reported the presence of an oxic microzone around seagrass root tissues during darkness. In the submerged freshwater macrophyte *Egeria densa*, the oxygen partial pressure was sustained in the root lacunae over 30 h after darkness (Sorrell and Dromgoole, 1987). Since the lacunae area in the rhizome and root tissues of *Thalassia testudinum* occupies over 50% of the cross-sectional area (Lee, unpublished data), stored oxygen in the lacunae of below-ground tissues may prevent toxic sulfide intrusion into root tissues during dark periods, when *T. testudinum* is illuminated sufficiently during daytime.

4.3. Sulfide dynamics with sediment depth

Seagrass below-ground biomass shows considerable vertical stratification within the sediments (Duarte et al., 1998). In the present study, most below-ground tissue biomass and surface area of *Thalassia testudinum* is located in the top 10 cm of the sediment. At sediment depths containing high below-ground biomass, pore water sulfide concentrations in *T. testudinum* beds decreased significantly during high photosynthetic periods. However, in deeper sediments (which contain little below-ground biomass) and at all sediment depths in adjacent bare areas, pore water sulfide concentrations did not show diel cycles. The pattern of pore water sulfide concentration changes with sediment depth reflects oxygen release from below-ground tissues which is correlated with below-ground tissue biomass and surface area.

Significantly higher sulfate reduction rates have been reported in seagrass beds than adjacent bare areas (Pollard and Moriarty, 1991; Isaksen and Finster, 1996; Holmer and Nielsen, 1997). In the present study, lower pore water sulfide concentrations in bare areas were probably results of the lower sulfate reduction rates in bare areas than in seagrass beds. The low sulfate reduction rates in bare areas are likely a result of low organic content of the sediment. Additionally, bare areas receive no exudation of labile organic matter from seagrass below-ground tissues. In bare areas pore water sulfide concentrations were lowest at the top 10 cm of the sediment where sediment organic C and N content were highest. Seagrass canopies can reduce current velocity on the sediment–water interface (Fonseca et al., 1982, 1983; Fonseca and Fisher, 1986; Gambi et al., 1990) and minimize water exchange between sediments and overlying water. Thus, current velocity and water exchanges between sediments and the water column are probably higher in the bare areas than in seagrass beds. High current velocities can push the anoxic layer deeper and decrease nutrient concentrations in pore waters (Koch, 1999). Therefore, considerably low sulfide concentrations at the top 10 cm of the sediment in bare areas were probably caused by high exchange rates between anoxic pore water and oxic overlying water at shallower sediment depths.

In conclusion, sediment pore water sulfide concentrations in *Thalassia testudinum* beds decreased significantly during peak photosynthetic periods in contrast to adjacent

bare areas, which did not exhibit any apparent diurnal variation. The decreases in pore water sulfide concentrations in seagrass beds during mid-day suggest sulfide oxidation occurs in the rhizosphere by photosynthetically produced oxygen that is transported from leaf tissues. Consequently, the sulfide oxidation likely prevents intrusion of toxic sulfide into plant tissues.

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References

- Allam, A.I., Hollis, J.P., 1972. Sulfide inhibition of oxidases in rice roots. *Phytopathology* 62, 634–639.
- Bagarinao, T., 1992. Sulfide as an environmental factor and toxicant: tolerance and adaptations in aquatic organisms. *Aquat. Toxicol.* 24, 21–62.
- Caffrey, J.M., Kemp, W.M., 1991. Seasonal and spatial patterns of oxygen production, respiration and root-rhizome release in *Potamogeton perfoliatus* L. and *Zostera marina* L. *Aquat. Bot.* 40, 109–128.
- Carlson, P.R., Yarbrow, L.A., Barber, T.R., 1994. Relationship of sediment sulfide to mortality of *Thalassia testudinum* in Florida Bay. *Bull. Mar. Sci.* 54, 733–746.
- Cline, J.D., 1969. Spectrophotometric determination of hydrogen sulfide in natural waters. *Limnol. Oceanogr.* 14, 454–458.
- Duarte, C.M., Merino, M., Agawin, N.S.R., Uri, J., Fortes, M.D., Gallegos, M.E., Marbá, N., Hemminga, M.A., 1998. Root production and belowground seagrass biomass. *Mar. Ecol. Prog. Ser.* 171, 97–108.
- Dunton, K.H., 1996. Photosynthetic production and biomass of the subtropical seagrass *Halodule wrightii* along an estuarine gradient. *Estuaries* 19, 436–447.
- Dunton, K.H., Tomasko, D.A., 1994. In situ photosynthesis in the seagrass *Halodule wrightii* in a hypersaline subtropical lagoon. *Mar. Ecol. Prog. Ser.* 107, 281–293.
- Fenchel, T., Bernard, C., 1995. Mats of colourless sulphur bacteria. I. Major microbial processes. *Mar. Ecol. Prog. Ser.* 128, 161–170.
- Flessa, H., 1994. Plant-induced changes in the redox potential of the rhizosphere of the submerged vascular macrophytes *Myriophyllum verticillatum* L. and *Ranunculus circinatus* L. *Aquat. Bot.* 47, 119–129.
- Fonseca, M.S., Fisher, J.S., 1986. A comparison of canopy friction and sediment movement between four species of seagrasses with reference to their ecology and restoration. *Mar. Ecol. Prog. Ser.* 29, 15–22.
- Fonseca, M.S., Fisher, J.S., Zieman, J.C., Thayer, G.W., 1982. Influence of the seagrass *Zostera marina* (L.) on current flow. *Estuar. Coastal Shelf Sci.* 15, 351–364.
- Fonseca, M.S., Zieman, J.C., Thayer, G.W., Fisher, J.S., 1983. The role of current velocity in structuring seagrass meadows. *Estuar. Coastal Shelf Sci.* 17, 367–380.
- Froelich, P.N., 1980. Analysis of organic carbon in marine sediments. *Limnol. Oceanogr.* 25, 564–572.
- Gambi, M.C., Nowell, A.R.M., Jumars, P.A., 1990. Flume observations on flow dynamics in *Zostera marina* (eelgrass) beds. *Mar. Ecol. Prog. Ser.* 61, 159–169.
- Goodman, J.L., Moore, K.A., Dennison, W.C., 1995. Photosynthetic responses of eelgrass (*Zostera marina* L.) to light and sediment sulfide in a shallow barrier island lagoon. *Aquat. Bot.* 50, 37–47.

- Gundersen, J.K., Jørgensen, B.B., Larsen, E., Jannasch, H.W., 1992. Mats of giant sulphur bacteria on deep-sea sediments due to fluctuating hydrothermal flow. *Nature* 360, 454–456.
- Hedges, J.I., Stern, J.H., 1984. Carbon and nitrogen determinations of carbonate-containing solids. *Limnol. Oceanogr.* 29, 657–663.
- Herzka, S.Z., Dunton, K.H., 1997. Seasonal photosynthetic patterns of the seagrass *Thalassia testudinum* in the western Gulf of Mexico. *Mar. Ecol. Prog. Ser.* 152, 103–117.
- Hines, M.E., Lyons, W.B., 1982. Biochemistry of nearshore Bermuda sediments. I. Sulfate reduction rates and nutrient generation. *Mar. Ecol. Prog. Ser.* 8, 87–94.
- Holmer, M., Nielsen, S.L., 1997. Sediment sulfur dynamics related to biomass-density patterns in *Zostera marina* (eelgrass) beds. *Mar. Ecol. Prog. Ser.* 146, 163–171.
- Ingold, A., Havill, D.C., 1984. The influence of sulphide on the distribution of higher plants in salt marshes. *J. Ecol.* 72, 1043–1054.
- Isaksen, M.F., Finster, K., 1996. Sulphate reduction in the root zone of the seagrass *Zostera noltii* on the intertidal flats of a coastal lagoon (Arcachon, France). *Mar. Ecol. Prog. Ser.* 137, 187–194.
- Jørgensen, B.B., 1977a. Bacterial sulfate reduction within reduced microniches of oxidized marine sediments. *Mar. Biol.* 41, 7–17.
- Jørgensen, B.B., 1977b. Distribution of colorless sulfur bacteria (*Beggiatoa* spp.) in a coastal marine sediment. *Mar. Biol.* 41, 19–28.
- Jørgensen, B.B., 1977c. The sulfur cycle of a coastal marine sediment (Limfjorden, Denmark). *Limnol. Oceanogr.* 22, 814–832.
- Jørgensen, B.B., 1982. Mineralization of organic matter in the sea bed — the role of sulfate reduction. *Nature* 296, 643–645.
- Jørgensen, B.B., Revsbech, N.P., 1983. Colorless sulfur bacteria, *Beggiatoa* spp. and *Thiovulum* spp. in O_2 and H_2S microgradients. *Appl. Environ. Microbiol.* 45, 1261–1270.
- Joshi, M.M., Hollis, J.P., 1977. Interaction of *Beggiatoa* and rice plants: detoxification of hydrogen sulfide in the rice rhizosphere. *Science* 195, 179–180.
- Koch, E.W., 1999. Preliminary evidence on the interdependent effect of currents and porewater geochemistry on *Thalassia testudinum* Banks ex König seedling. *Aquat. Bot.* 63, 95–102.
- Koch, M.S., Mendelssohn, I.A., 1989. Sulphide as a soil phytotoxin: differential responses in two marsh species. *J. Ecol.* 77, 565–578.
- Koch, M.S., Mendelssohn, I.A., McKee, K.L., 1990. Mechanism for the hydrogen sulfide-induced growth limitation in wetland macrophytes. *Limnol. Oceanogr.* 35, 399–408.
- Lee, R.W., Kraus, D.W., Doeller, J.E., 1999. Oxidation of sulfide by *Spartina alterniflora* roots. *Limnol. Oceanogr.* 44, 1155–1159.
- Nelson, D.C., Jørgensen, B.B., Revsbech, N.P., 1986. Growth pattern and yield of a chemoautotrophic *Beggiatoa* sp. in oxygen-sulfide microgradients. *Appl. Environ. Microbiol.* 52, 225–233.
- Oremland, R.S., Taylor, B.F., 1977. Diurnal fluctuations of O_2 , N_2 , and CH_4 in the rhizosphere of *Thalassia testudinum*. *Limnol. Oceanogr.* 22, 566–570.
- Pedersen, O., Borum, J., Duarte, C.M., Fortes, M.D., 1998. Oxygen dynamics in the rhizosphere of *Cymodocea rotundata*. *Mar. Ecol. Prog. Ser.* 169, 283–288.
- Penhale, P.A., Wetzel, R.G., 1983. Structural and functional adaptations of eelgrass (*Zostera marina* L.) to the anaerobic sediment environment. *Can. J. Bot.* 61, 1421–1428.
- Pollard, P.C., Moriarty, D.J.W., 1991. Organic carbon decomposition, primary and bacterial productivity, and sulphate reduction, in tropical seagrass beds of the Gulf of Carpentaria, Australia. *Mar. Ecol. Prog. Ser.* 69, 149–159.
- Pulich, Jr. W.M., 1983. Growth response of *Halophila engelmanni* Aschers. to sulfide, copper, and organic nitrogen in marine sediments. *Plant Physiol.* 71, 975–978.
- Roberts, D.G., Caperon, J., 1986. Lacunar gas discharge as a measure of photosynthesis in seagrasses. *Mar. Ecol. Prog. Ser.* 29, 23–27.
- Roberts, D.G., McComb, A.J., Kuo, J., 1984. The structure and continuity of the lacunar system of the seagrass *Halophila ovalis* (R. Br.) Hook F. (Hydrocharitaceae). *Aquat. Bot.* 18, 377–388.
- Sand-Jensen, K., Prahl, C., Stokholm, H., 1982. Oxygen release from roots of submerged aquatic macrophytes. *Oikos* 38, 349–354.

- SAS Institute, 1989. SAS/STAT Guide for Personal Computers, Version 6, SAS Institute, Cary, NC, 4th Edition.
- Smith, R.D., Dennison, W.C., Alberte, R.S., 1984. Role of seagrass photosynthesis in root aerobic processes. *Plant Physiol.* 74, 1055–1058.
- Sorrell, B.K., 1994. Airspace structure and mathematical modelling of oxygen diffusion, aeration and anoxia in *Eleocharis sphacelata* R. Br. roots. *Aust. J. Mar. Freshwat. Res.* 45, 1529–1541.
- Sorrell, B.K., Dromgoole, F.I., 1987. Oxygen transport in the submerged freshwater macrophyte *Egeria densa* Planch. I. Oxygen production, storage and release. *Aquat. Bot.* 28, 63–80.
- Sorrell, B.K., Dromgoole, F.I., 1988. Oxygen transport in the submerged freshwater macrophyte *Egeria densa* Planch. II. Role of lacunar gas pressures. *Aquat. Bot.* 31, 93–106.
- Thode-Andersen, S., Jørgensen, B.B., 1989. Sulfate reduction and the formation of ^{35}S -labeled FeS , FeS_2 , and S^0 in coastal marine sediments. *Limnol. Oceanogr.* 34, 793–806.
- Thursby, G.B., 1984. Root-exuded oxygen in the aquatic angiosperm *Ruppia maritima*. *Mar. Ecol. Prog. Ser.* 16, 303–305.
- Van Wijck, C., De Groot, C.-J., Grillas, P., 1992. The effect of anaerobic sediment on the growth of *Potamogeton pectinatus* L.: the role of organic matter, sulphide and ferrous iron. *Aquat. Bot.* 44, 31–49.
- Visser, J.M., Robertson, L.A., van Verseveld, H.W., Kuenen, J.G., 1997. Sulfur production by obligately chemolithoautotrophic *Thiobacillus* species. *Appl. Environ. Microbiol.* 63, 2300–2305.
- Zimmermann, C.F., Price, M.T., Montgomery, J.R., 1978. A comparison of ceramic and teflon in situ samplers for nutrient pore water determinations. *Estuar. Coastal Mar. Sci.* 7, 93–97.